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Literature Review:

Business Applications of Blockchain Technology and Hyperledger Framework to
Maintain Vehicular Depreciation and Ownership History

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Kaustubh Dutta
Master's in E-Business Technologies
University of Ottawa, ON, Canada
Student Number: 300067370

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Kaustubh Dutta

University of Ottawa

Author Note

Kaustubh Dutta, Department of Engineering, University of Ottawa (Master's in E-Business Technologies).

This literature review highlights the Business Applications of Blockchain technology and proposes the uses case for Hyperledger Framework.

Correspondence concerning this article should be addressed to Kaustubh Dutta, Department of Engineering, University of Ottawa, 75 Laurier Avenue E, Ottawa, ON K1N 6N5.

Contact: kdutt075@uottawa.ca

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Abstract

Blockchain technology is gaining gradual importance in the way transactions would be performed in the near future. Here transactions are not limited to banking, financial platform, insurance domain or the practice of using transactional cards, but the transfer of information in a shared ecosystem in a variety of application areas. Nayak and Dutta (2017), highlights the work of Satoshi Nakamoto who was the first person to conceptualize Blockchain back in 2008, "a new electronic cash system" (Bitcoin cryptocurrency) where transaction could take place between more than one person or a participating body in a complete peer to peer protocol without the requirement of a trusted third party. Using the concept of Blockchain, the advantages of this application is felt as a requirement in every technological domain. Wherever there is a transfer of information (transfer of funds, movement of goods through the supply chain or any kind of contractual agreements) can be leveraged by using a Blockchain Application Network.

Problem Statement: Blockchain Technology has many advantages which is seen through its application in various sectors as discussed in the literature review, however the resistance to accept it as a major technological break-through finds it difficult to be accepted for the progress of this new generation of technology.

Significance: The growth of commerce and trade has led to the creation of disparate ledger system that is vulnerable to errors, fraud and misinterpretation. Blockchain provides transparent transactions through a secured and synchronised record of transactions by smoothening capital flow, speeds processes, lowering transaction costs through security and trust.

Context: With the internet reaching every corner of the world and the trend of growing faster data connection, most of the processes are going digital and getting connected. It is important to have a unique platform that would be transparent, secured, tamper proof and free from the intervention of a centralized third party governing the processes which can be made possible by Blockchain.

Objective: The objective of this literature review is to analyse the content in the research articles that have been studied and to highlight the trend in the applications of Blockchain technology as well as to identify the gaps. As per the studies, Blockchain technology has a significant potential to be a game changer in the various technological domains and application areas, so the formulation of a clear logical walkthrough needs to be set in terms of the key areas of strengths and weaknesses. The research studies have been done in recent time from 2015 to 2018 by considering the various technologies associated with Blockchain, like Hyperledger (focussed mainly) and Ethereum in the fields of healthcare, smart cities & IOT, automotive industry, smart contracts and data protection and privacy.

Keywords: distributed consensus ledger, Hyperledger Fabric, Smart Contract, Automotive sector.

Introduction

Yuan and Wang (2016) says Blockchain technology has evolved in the adaptability in recent times in a variety of application areas depending upon the incorporation of the abilities specific to operations. Christidis and Devetsikiotis (2016), examines that the financial sector was the major focus for Blockchain Technology to grow its significance however the benefits has moved to a wide array of applications. With each sector demanding accuracy of process, transparency in record maintenance and faster processing of data, the need for Blockchain Technology has grown to a significant demand which has created a technological gap between the requirements and the implementation. As the technology is moving forward, the requirement of a shared system has become a necessity due to the practicality and convenience. Biswas and Muthukumarasamy (2016) states that with this kind of an approach, it would be a backbone for the concept of smart cities that are getting conceptualized, where it is required that the information technology to have an interconnected and integrated management with the physical, social and business infrastructures that would provide better services for an efficient and optimal utilization of available resources. Internet of Things (IoT), cloud computing, interconnected networks and smart cities have positive capabilities to drive innovative solutions through a more direct interaction between the people and the government. Despite having benefits, there would be a digital disruption and pose challenges related to information security and privacy (Biswas & Muthukumarasamy, 2016). With the massive data that is expected to be generated as per the technological trends and an approach in the shift of current business processes by seeing the benefits of Big Data, the need for a distributed peer to peer network involving the interaction of non-trusting members without a trusted intermediary is gaining importance. The automation of the multi-chain process by using smart contracts is gradually gaining importance as we move into the blockchain IoT combination: 1) facilitating the sharing of services and the resources, thereby creating a shared marketplace of services and 2) achieving automation in a cryptographical verifiable manner for the complete ecosystem (Christidis & Devetsikiotis, 2016). With all the approaches in mind, the requirement for the processing of shared data, there has been increased concerns for surveillance, security breaches, privacy of personal information as there has always been a third party that controls the transfer of information. This concern gives an added advantage to Blockchain technology as it removes the requirement and presence of a central authority governing the transfer of information based on content, sender and recipient. Research carried out by (Zyskind, Nathan, & Pentland, 2015) states in their work that the new approach works on a decentralized network of peers accompanied by a public ledger where each transaction taking place in the blockchain network is verified by all the members, part of the blockchain. As a result, this acts as an automated access-control manager that does not require a trust in a third party, but the complete process takes place in an efficient and fast end to end process by a verified automated approach.

As a result, the applications of a Blockchain technology is rising and gaining importance. People have started trusting the new technology as it is gaining confidence from a concept to a reality as this approach is still very new. In this literature review, various application areas have been touched where Blockchain technology has the capability of improving existing frameworks with the up gradations in technology that is continuously taking place. We do keep in mind, that this technology has received initial setbacks in terms of not being accepted in various parts of the

world due to the rise and fall of bitcoin crypto-currency values and issues surrounding it, however the technology has a significant potential of disrupting the majority chunk of various processes in different sets of application fields with a central agenda of eliminating the requirement of a central authority in the business processes (Nayak & Dutta, 2017). The applications of Blockchain are highlighted in this paper, in the fields of: smart cities, healthcare, ticketing service, smart contracts for Internet of Things (IoT), intelligent transportation and automative security and privacy, food and nutrition sector. Also the benefits of Hyperledger fabric in Blockchain technology and its advantages for small and medium scale business enterprises.

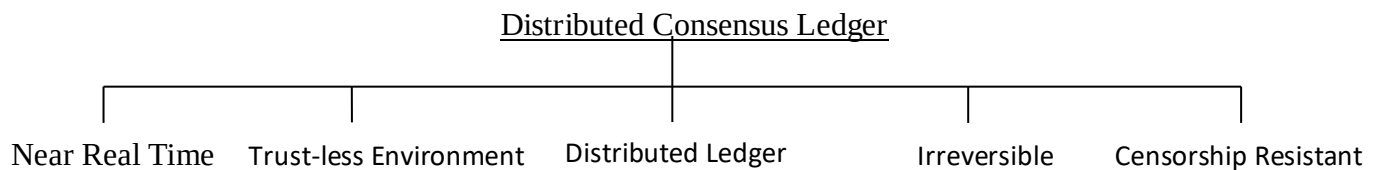


Fig 1: Properties of Blockchain

Source: Deloitte Analysis, 2018

Related Work

There are few studies, that have been analysed and focused primarily on the security and data protection, that the technology of Blockchain provides (Nayak & Dutta, 2017). The problem of privacy is a great concern, as significant amount of related data is transferred and passed along with the actual transactions during a normal day to day activities (e.g.- personal details, banking information, etc). The data ownership, transparency, auditability and fine grained access control is among the key highlights according to Zyskind et al., 2015 that makes this technology unique and popular in comparison to a standard distributed ledger. Along with the data privacy features, the functionality of access between the user and service in a particular decentralized platform has been shown which is in fact unique and quite exceptional.

Through a systematic approach, this literature review brings out the application of the Blockchain technology, in the field of healthcare. In many countries, the access to basic healthcare challenge due to a variety of unexpected factors. According to Mettler (2016), applications including the concept of a smart health care management, identifying the circulation of counterfeit drugs and medicines as well as a seamless accessibility to insurance facilities would be beneficial and a boon to the society. (Zhang, Xue, & Huang, 2017) says that network based healthcare, which makes the use of the data collected by sensors and facilitating the collaboration through the decisions taken by participating bodies is innovative and quite rewarding.

The need for an innovative, intelligent and a smart transportation system, is in fact a global requirement due to the continuous rise in operational and infrastructure costs along with the increase in accidents. The key insights as identified by (Yuan & Wang, 2016) in their research

paper brings out the new concept where vehicles, traffic lights, off-shore control centres and a variety of additional supporting functionalities to be included in a consensus mechanism all interacting with each other. It brings out the additional features of transport, like ride sharing along with the emphasis of safety which is a significant breakthrough (Dorri, Steger, & Jurdak, 2017) .

With the concept of Internet of Things (IoT), as stated in the work of (Christidis & Devetsikiotis, 2016) there would be a massive requirement of data sharing to take place with a high demand of continuous interactions and expectations of a seamless process flow. It has been identified through the research, that with the realization of IoT and Big Data platform, the presence of a central authority would hamper the smooth and efficient flow of data between different parties as there would be a protocol mismatch (Biswas & Muthukkumarasamy, 2016). (Dorri, Kanhere, Jurdak, & Gauravaram, 2017) the Blockchain based approach solves this issue due to the concept of a shared architecture with a tamper proof approach. It would be simple and easy to identify and track each transaction or technical logs that gets created (Ali, Ali, Musa, & Zahrani, 2017).

In general, Blockchain finds many applications that is discussed in detail in due course of the literature review. According to Deloitte (2018), as per a recent study carried out, clearly highlights the advantages of having Blockchain for the benefit of the public sector as it would help in various intergovernmental transactions and creating a unified model of trust. Along with that, the most important concept is the use of Hyperledger technology of Blockchain that has the capability of scaling small- and large-scale business enterprises.

Research Design

The literature review was carried out by studying content from a variety of sources. There were initial searches made in various databases and content of all sorts of articles and published work were studied. Extensive databases were considered for research, journals, conference papers and grey articles. The type of literature to be reviewed was finalized by executing a variety of search queries to match specific results related to Blockchain and associated technologies with specific applications. Databases that have been used are Scopus, Web of Science, IEEE Xplore, Science Direct, Google Scholar and ACM Digital library for the literature review.

<u>Digital Library</u>	<u>Search Queries</u>
Scopus	• Blockchain and healthcare • Blockchain and automotive security
Web of Science	• Hyperledger and IOT

IEEE Xplore	• Blockchain and Internet of Things in Hyperledger • Blockchain and Transportation • Smart City or Blockchain • Blockchain and Food/Nutrition Sector • Risk of Hyperledger Fabric Smart Contracts
Science Direct	• Hyperledger fabric
Google Scholar	• Decentralizing Privacy and Protect Personal Data
ACM Digital Library	• Hyperledger or scalable or industrial blockchain

Literature Review on Business Applications of Blockchain Technology

In this section of literature review applications of Blockchain Technology in various fields have been discussed.

Privacy and Data Protection

With the increase of data usage across shared platforms it has become a major concern for the privacy and protection of data. As discussed by (Nayak & Dutta, 2017) there are continuous challenges of data breaches that take place both due to negligence or ignorant exposures to vulnerable threats. With the increase in technology, more is the risk of such threats as systems get complex by opening the un-monitored vulnerable points. Blockchain transactions have a signified advantage due to the decentralization of trust as it eliminates the use of a central authority in governing the decisions. Moreover, the cryptographic hash is almost immutable and impossible to break as slight modifications would notify all the members in the blockchain with a potential risk of invalidating any unmonitored transactions trying to take place. As said by (Zyskind et al., 2015) the contributions made in constructing a personal data management platform through a vital source of trusted computing. As most of the applications that run on our computers and smart-phones collect personal information, so it is important to have: data ownership, transparency and fine grained access control. Zyskind et al., 2015 proposes a solution of an end user specific protocol

implemented through Blockchain technology where data transactions that govern read/write operations would be selective in nature through *Check Policy* where the user authorized to a particular service would be able to access the data intended for his use only.

Nayak & Dutta, 2017 proposed the future extension of the application of Blockchain in the fields of storage and processing. Sometimes it is a challenge when raw data needs to be shared to third parties, but valuable data gets leaked out. This application has been approached by an application of a mobile based app used for voting where the results are stored in a public ledger where no one knows about the individual votes, but everyone can see the election results (Zyskind et al., 2015). This application finds a suitable solution in countries where elections results are rigged and pre-decided.

So personal and sensitive data like governmental and financial information can be effectively stored in public ledger without compromising on security and the threat to data breach. The decentralized concept of Blockchain makes it simple, yet efficient and powerful for making legal and regulatory decisions. Depending upon the requirements, laws and regulations can be programmed into the Blockchain for highly secured and tamper proof implementation (Nayak & Dutta, 2017).

Applications of Blockchain Technology in the automotive and transportation sectors with the concept of Smart Technology and Internet of Things (IoT):

Intelligent Transportation System oriented Blockchain Model

According to (Yuan & Wang, 2016), the implementation of Blockchain, like that of an OSI model in the transportation sector is an interesting concept. Here in this paper, a seven-layer conceptual model has been presented for a standardized Blockchain architecture system for the implementation of an intelligent transportation.

Physical layer: The Physical layer comprises of (vehicles, sensor based devices, traffic lights, on board devices, roadside units, GPS modules, driver's smartphones etc.) which generates data and can be securely recorded into the immutable and irrevocable ledger of blockchain. The sensor placed in the chips can be used as a mining platform for establishing a connection to the blockchain.

Data layer: The Data layer is responsible for providing asymmetric encryption, time stamping, governing of the hash algorithms in a chronological order of each blocks of the Blockchain.

Network layer: Responsible for data forwarding, mechanisms for distributed networking and verification.

Consensus layer: Implementation of the key concept of Blockchain, by promoting decentralized nodes for reaching consensus on data validity to develop the foundation for mutual trust.

Incentive layer: Implementation of an economic reward system that would generate bitcoins on creation of a block to motivate the network for the continuous efforts of data verification.

Contract layer: This layer serves as platform for self-executing smart contracts, algorithms and scripting mechanisms depending on the business requirements of the intelligent transportation system.

Application Layer: Ride-sharing, Logistics, Asset Mangement....
Contract Layer: Algorithm, Mechansim, Smart Contracts
Incentive Layer: Issuance, Allocation
Consensus Layer: PoW, PoS, DPos
Network Layer: P2P Network, Data Forwarding, Data Verification
Data layer: Data blocks, Time Stamps, Hash Functions, Encryption....
Physical layer: Devices, Vehicles, Assets

Fig 1: ITS Oriented Blockchain Model

The above model has additional features like that of ride-sharing, fleet management that can be seen through the application layer (Yuan & Wang, 2016).

Automotive Security and Privacy

With the implementation of Blockchain technology, there are various untouched areas that can be covered due to the sharing of information via a secured channel. Automotive security based on a public and private key pair for monitoring of vehicular status finds application in insurance, smart charging services, ride sharing (Dorri et al., 2017). The benefit for an open insurance policy can be achieved where an insurance company continuously monitors the status of a vehicle and the annual premium value can be set depending on the car condition as accidents can be reported to the insurance company's cloud storage automatically via the Blockchain technology. The concept of ride sharing services by cab aggregators and smart charging services would be a new concept as it would be environment friendly by evaluating the demand between riders requesting a cab and the availability of drivers by utilizing the shared data (Dorri et al., 2017) and optimize the requests depending upon the time of the day, remaining fuel in car and distance to travel.

Winning Applications in the Public Sector

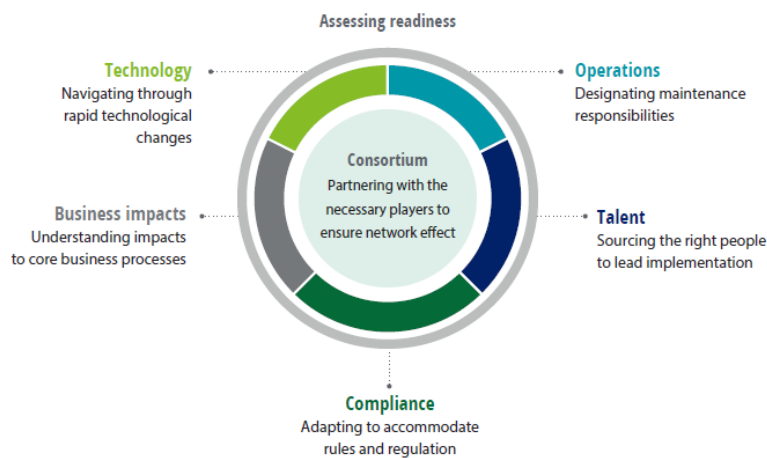


Fig 2: Six Factors that would create a successful Blockchain application in the public sector

Source: Adapted from Eric Piscini et al., Taking Blockchain live: The 20 questions that must be answered to move beyond proof of concept, Deloitte 2017

Blockchain has a potential of creating a unique ecosystem to provide a specific solution in the public sector to meet the gap. This would help this technology to be accepted and adapted as: Consortium (Collaboration with the required parties), Operation, Business Impacts, Talent, Compliance (abiding to the rules and laws) and Technology (Deloitte, 2018). The use case of Intergovernmental transaction is a perfect example as there is a bridge in the type of transactions that are taking place where it has been noted, that it results in errors in consolidated financial statements and as per the report issued by US Government Accountability Office, that there are unaccounted losses in billions of dollars due to errors and low trust in reporting. Looking at the desirability of the new technology that would provide a solution through an accurate record of transactional reporting with a feasibility of business transactions that would ensure security to be the key points that could benefit governments in general, at large. Since this technology adds transparency through a low trust process would add significant value to the Intergovernmental transactions with interagency agreement (IAA) into a "smart-contract" of the Blockchain.

Deloitte, 2018 claims that the potential solutions are gradually becoming popular and it offers government organisations an opportunity for true collaborations and transformation.

Secured Internet of Things (IoT)

The technology of Blockchain and Smart Contracts for Internet of Things (IoT) enables the process of the scripts that reside on the Blockchain (Smart Contracts) which allow for the automation of a multi-step processes. The combination of both IoT and Blockchain which would facilitate the sharing of services and allowing to automate in a cryptographic verifiable manner

(Christidis & Devetsikiotis, 2016). With the Blockchain in place the decentralized operation without the need to have a central authority and achieve the same functionality as that of before but by minimizing the risk of shared data of IoT. Blockchain enables a shift from the massive expansive IoT through a sustainable approach (Ali et al., 2017).

Smart Homes

The application of Blockchain can help mitigate the issues related to IoT Security and Privacy for Smart Homes. The major challenge for this issue is due to the massive scale and distributed nature of the IoT networks. The approach for the IoT network is to address three main tiers: cloud storage, overlay and smart home. Here the concept of Bitcoin that works on a changeable Public Key (PK) can be incorporated to decrease the network overhead delay by grouping clusters for each node of the IoT network (Dorri et al., 2017). This overlay helps in the realization of Blockchain in conjugation with two key lists: requester key and requestee key. Qualitative arguments that a smart home tier achieves confidentiality, integrity and availability of integral data and thereby minimising the risk of Distributed Denial of Service (DDoS) attack. The architecture serves as a strong pillar with a concept of a local block chain which can be used to keep track of local transactions as well as Home Miner that centrally processes incoming and outgoing transactions. The Blockchain based Smart Home incorporates a transaction handling procedure by using the concept of shared key between any two device (external or internal) as the keys are valid which ensures high security (Dorri et al., 2017).

Smart Cities

The system architecture is getting complex for the gradual achievement of Internet of Things (IoT) and the demand for delivering innovative solutions, direct interaction & collaboration between people and government are taking place at a large scale. Information security and privacy is posing a posing a great threat which Blockchain Technology can fulfil through a smart communication platform. Biswas & Muthukkumarasamy, 2016 proposed a security framework that clearly highlights the benefits of Blockchain Technology as it would never allow an attacker to compromise 51% of the system to surpass the hashing power of the target network as the blocks within the system work on complex verification algorithms to allow a transaction to be validated and added to the chain.

This identifies for the creation of a common platform that would enable for all devices to communicate securely in a distributed environment for a smart city that would improve on the reliability, fault tolerance, and efficiency in operation along with the required scalability.

Hyperledger and its potential to scale up small, medium and large-scale enterprise

The three types of top-enterprise Blockchain are Quorum, Hyperledger Fabric and Corda. As per Linux Foundations 2017, Hyperledger Fabric is an open source, collaborative approach that ensures transparency, longevity, interoperability and support to bring the blockchain technologies forward to mainstream commercial application and is supported by the Linux Foundations since December 2015. According to (Yang, Zhihui, & Wu 2018) Hyperledger fabric works on the principle of two roles for a peer - endorser and committer. The concept of endorser (preventing unstable transactions in the network) and committer (facilitates validation in the ledger) makes possible of secured transactions and developing trust. This helps in satisfying business requirements and delivering value to customers (Johng, Kim, Hill & Chung 2018) as it provides a sense of confidence and transparency in the business processes working in a digital platform. Hyperledger Fabric makes the implementation simple as specific rules or smart contracts get activated in between specific components of a business processes like digital assets, participants and the transactions as stated by Kaid and Eljazzar (2018) which makes a smoother process flow in between supply chain parties.

Looking at the value that Hyperledger Fabric provides, Blockchain technology built on a similar kind of enterprise can scale up businesses and process flow model (Sousa, Bessani, & Vukolic, 2018).

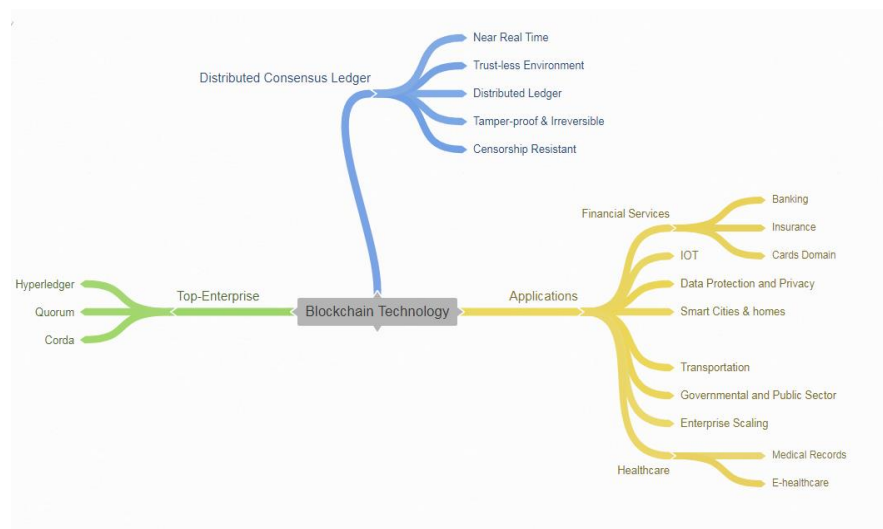


Fig 3: Mindmap of Blockchain Technology

Type of Scientific Research and Methodology:

- **Exploratory Research:** Blockchain is still in the research design phase so I would like to explore the key concepts and design of a prototype for the Blockchain Hyperledger Framework.
- **Research Method:** Qualitative (non- statistical analysis)

Differences between Ethereum and Hyperledger Blockchain Platforms:

<u>Ethereum</u>	<u>Hyperledger</u>
Decentralized platform that runs on Smart Contracts.	Hyperledger is a multi-project open source collaborative effort hosted by The Linux Foundation.
Public Network: Everyone can connect with an internet and basic privileges and permissions.	Consortium Network: Complete privacy and confidentiality.
Crypto coin- Ether: Network specific coin that pays for transactions mined by miners.	No Crypto coin or token: Freedom to Blockchain developers if required.
Consensus Mechanism: • Proof of Work • # function is used for transactions • Conclusions are verified by other participants in the system and rewarded by newly generated “Ether”.	Usually No Consensus Mechanism- Byzantine Fault Tolerant System • Information is stored in a node. • Each node contains a status and on receiving an information. • The information is shared with the network and then updated.
Smart Contract: Solidity • Solidity is a contract-oriented, high-level language for implementing smart contracts designed to target the Ethereum Virtual Machine (EVM).	Smart Contract: Chaincode • Installed Smart Contracts- business logic is installed on the validators in the network before the network is launched. • On-chain smart contracts- deployment of the business logic as a transaction committed to the blockchain and then called by subsequent transactions.

Maintenance: Ethereum Developer Community	Maintenance: Linux Foundation
Type of Business used: Business to Customer (B2C)	Type of Business used: - Business to Business (B2B) - Confidential use cases

Brief overview of the Hyperledger Architecture :

- **Consensus Layer-** This layer is responsible for confirming the correctness of the set of transactions and generating an agreement on the order that constitute a block.
- **Smart Contract Layer-** The layer executes a business logic to check the validity for transaction requests to be processed.
- **Communication Layer-** Nodes that participate in a shared ledger instance communicate between each other through a peer-to-peer message transport protocol.
- **Data Store Abstraction-** Allows different data-stores to be used by other modules.
- **Crypto Abstraction-** Crypto algorithms or modules can be injected or scraped without affecting other modules.
- **Identity Services-** This is a type of a root of trust during the creation and setup of a blockchain instance: enrollment, registration of identities or system entities during network operation, and provides authentication, authorization and revocations.
 - **Policy Services-** Responsible for incorporating different policies specified in the System like endorsement, group management or consensus.
 - **APIs-** Applications and client's interface to blockchain applications using API's.
 - **Interoperation-** Interoperability between various blockchain instances.

Types of Hyperledger Frameworks:

Hyperledger Framework	Description	Smart Contract Support
Hyperledger Burrow	Permissionable smart contract machine (EVM).	✓
Hyperledger Fabric	Permissioned with channel support.	✓
Hyperledger Grid	Web Assembly-based project for building supply chain solutions.	X
Hyperledger Indy	Decentralized Identity.	X

Hyperledger Iroha	Mobile Application Focus.	✓
Hyperledger Sawtooth	Permissioned & permissionless support and EVM transaction family.	✓

Tools available for Hyperledger:

Tools	Description
Hyperledger Aries	Infrastructure for peer to peer interactions.
Hyperledger Caliper	Blockchain framework benchmark platform.
Hyperledger Cello	As a service deployment.
Hyperledger Composer	Model and build Blockchain Networks.
Hyperledger Explorer	View and explore data on the Blockchain.
Hyperledger Quilt	Ledger and Interoperability.
Hyperledger Transact	Advanced transaction execution and state management.
Hyperledger URSA	Shared Cryptographic Library.

Processing of Smart Contracts in Hyperledger Blockchain Framework

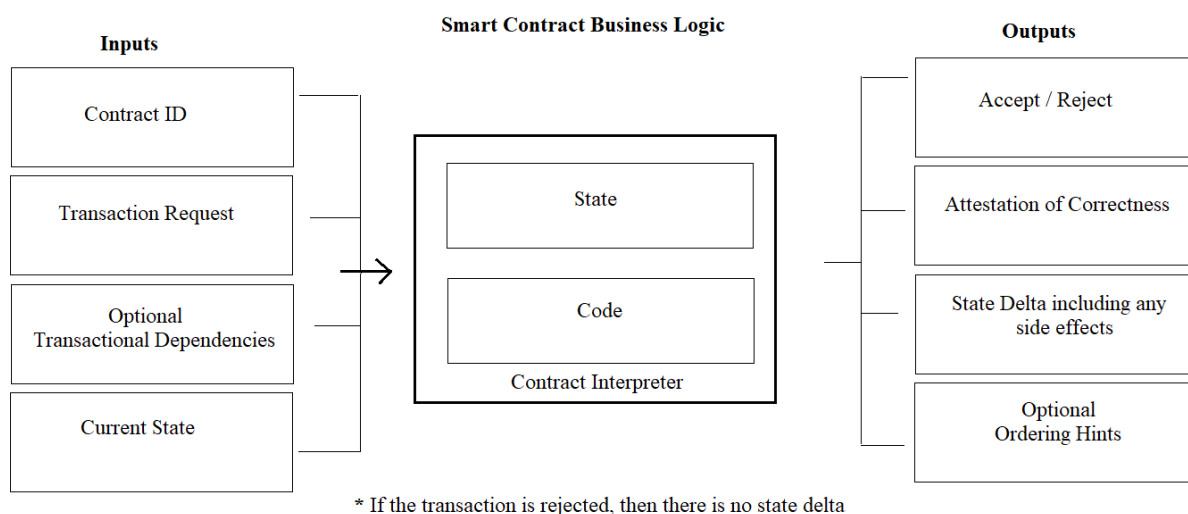


Fig 4: Processing of requests via Hyperledger Smart Contracts

Inputs to the Smart Contract System are: the contract identifier, the transaction request, any dependencies that may exist, and the current state of the ledger. The contract interpreter which is the middle block contains the current state of the ledger and the smart contract code. When the contract interpreter receives a request, it immediately checks and then rejects any invalid requests.

The required output is generated if the request is valid and accepted. The output includes a new state or any side effects. When all the stages of processing are complete, the interpreter packages the new state, an attestation of correctness, and any required ordering hints for the consensus services. That package is sent to the consensus service for final commitment to the blockchain network. The smart contract layer validates each request by ensuring that it conforms to policy and the contract specified for the transaction. Invalid requests are rejected and may be dropped from inclusion within a block, depending on the framework (Linux Foundations, 2018).

Two categories of validation errors: Syntax errors and Logical error.

For syntax errors such as invalid inputs, unverifiable signature, or any repeated transaction (due to errors or replay attacks) it should be ensured that the request is dropped. Logical errors are more complex, and the decision whether to continue the processing should be completely policy-driven. For example, if a request would result in double-spend or versioning control failure, you might want to log that request for auditing, if the policy requires to do so. The result of validating a request in a smart contract is a transaction that captures the transition to the new state. Each Hyperledger framework represents the state transition differently; for example, Hyperledger Fabric uses *change sets*, while Hyperledger Sawtooth uses cryptographically *verifiable states*.

The separation between *contract execution* and *consensus* complicates the commitment to the blockchain. For example, two requests executed in parallel can result in, inconsistencies in the contract state unless they are ordered in a specific way. In general, the consensus layer handles ordering. However, the smart contract has the option to provide hints about properties like ordering. Processing a request can generate side effects: results that are not captured in the state delta. Side effects can include event notifications, requests to other smart contracts, modifications to global resources, or even irreversible actions like releasing sensitive information or shipping a package (Linux Foundations, 2018).

Architectural Layers involved in the Interactions of Smart Contracts:

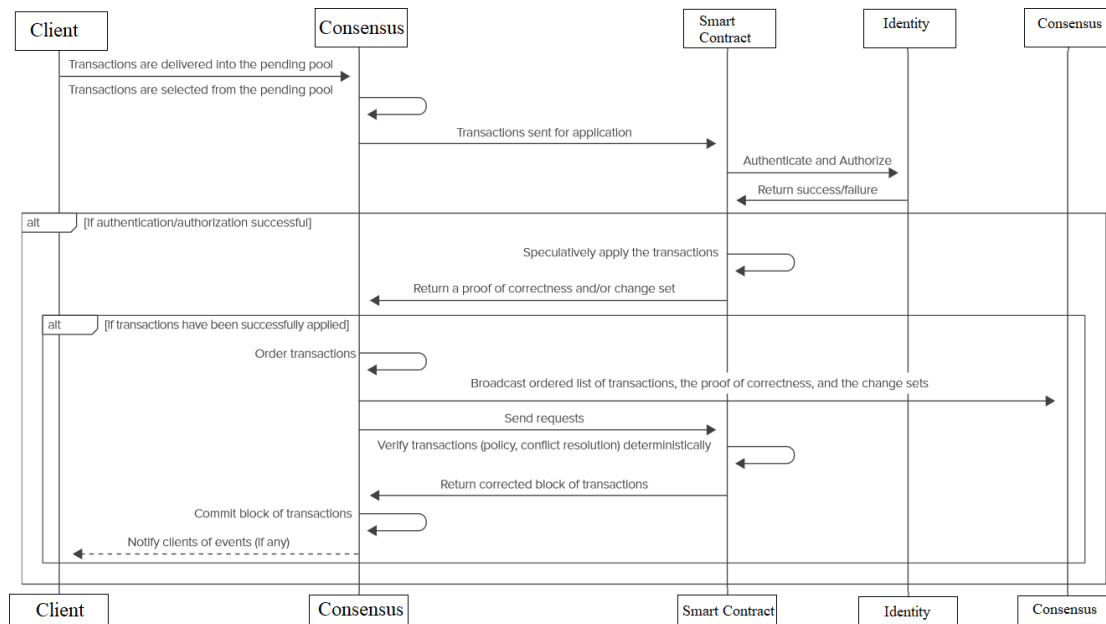


Fig 5: Interaction of Hyperledger Smart Contracts with other Architectural Layers

The smart contract layer works almost in hand in hand with the consensus layer. The smart contract layer accepts a proposal from the consensus layer. This proposal specifies the contract to be executed, the details of the transaction including the identity and credentials of the entity asking to execute the contract, and any other transactional dependencies (Linux Foundations, 2018).

The current state of the ledger and input from the consensus layer is used by the smart contract layer to validate the transaction.

The smart contract layer uses the identity services layer to authenticate and authorize the entity asking to execute the smart contract while processing transactions. Here two parameters are ensured: that the presence of the entity is validated and known on the blockchain network, and that the entity has the appropriate access to execute the smart contract. Identity can be granted through various methods: simple key-based identities, identities and credentials managed through the ledger, anonymous credentials, or managed identity services from an external certificate authority. After processing the transaction, the smart contract layer returns the state of the transaction whether it was accepted or rejected. For transactions states as accepted, the smart contract layer also returns an attestation of correctness, a state delta, and any optional ordering hints needed to ensure the transaction dependencies are considered (Linux Foundations, 2018).

The state delta includes the change sets and any side effects that should take place when the transaction is successfully committed by the peers.

Demonstration of Automotive Depreciation and Vehicular Ownership Use Case:

Tool Used: Hyperledger Composer

Hyperledger Composer is an extensive, open development toolset and framework to make developing blockchain applications easier. Hyperledger Composer supports the existing Hyperledger Fabric blockchain infrastructure and runtime, which supports blockchain consensus protocols to ensure that transactions are validated according to policy by the designated business network participants (Linux Foundations, 2018).

Block diagram of the prototype:

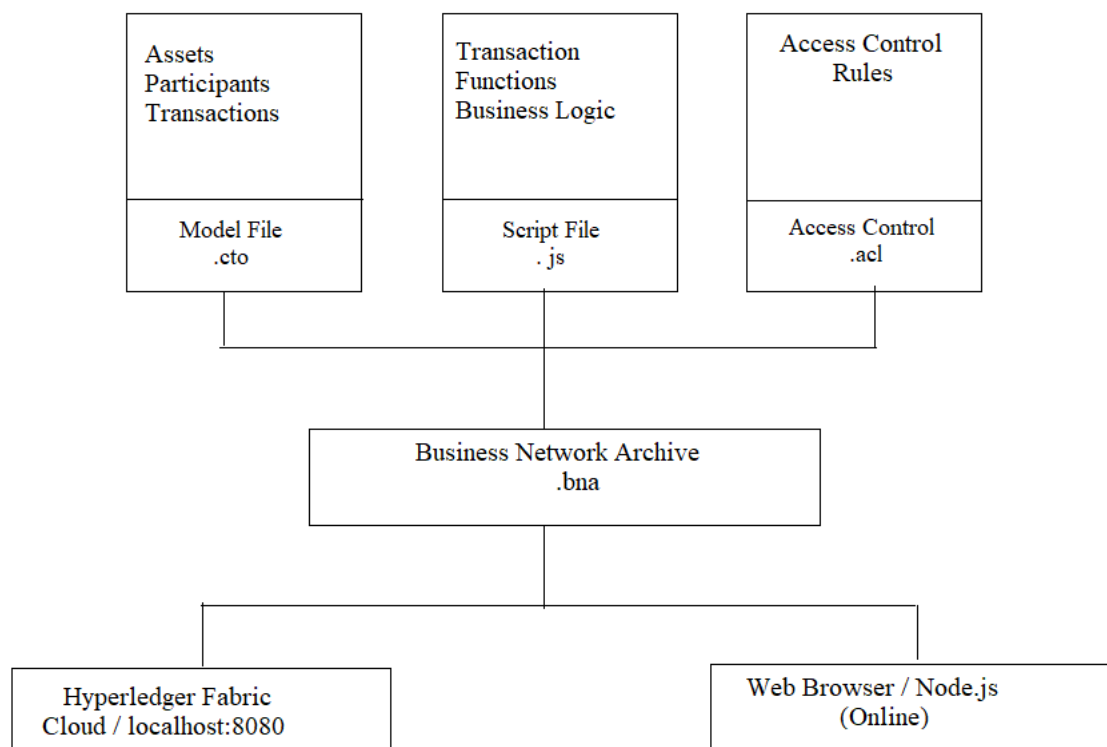


Fig 6: Architecture of the Prototype Use Case

Components part of the Hyperledger Blockchain Prototype:

<u>Component</u>	<u>Description</u>	
Participant	Manufacturers- Car Makers need to be registered in the Blockchain Network to prevent any presence of black listed companies in the car industry.	Traders- Any person eligible to make transactions (buy or sell) vehicles would be listed in the network.
Commodity	As soon as a vehicle gets rolled out of the factory is listed in the Hyperledger network.	
Transaction	Transfer of ownership that generates a tamper proof transaction ID and time stamp. 1. Manufacturers to Traders (vehicle rolling out of factory for the first time) 2. Between traders (subsequent transactions)	
Smart Contract	Depreciation in the price of SUV, Sedan, Car and Truck segment based on distance driven and ownership ranking.	

Details of Smart Contract:

1. In every transaction there would be a **depreciation of vehicular price** by % as mentioned in the Depreciation % table.
2. **Self-Transaction Protection:** No person can sell his/her vehicle to himself/herself to avail the concept of depreciation to reduce his/her vehicular insurance.
3. **Maximum number of transactions** of a vehicle is decided as per the smart contract beyond which no vehicle can be further transacted and will obtain an invalid status.
SUV-10 | Sedan- 5 | Hatch Back -3
4. **Ownership Ranking** is embedded with each car details to prevent fraud and to make a new owner aware regarding his/her ownership ranking.
5. **Ownership History-** Details of all previous ownership details are embedded in the commodity details with each passing transaction of a commodity.
6. Transaction Table:

SL NO	Vehicle Type	Maximum Number of Transactions allowed	Ownership Ranking that will allot invalid status	Comments
1.	SUV	10	10	After the 10th transaction (ownership ranking- 10) the vehicle will obtain an invalid

				status and the commodity cannot be further traded.
2.	Sedan	5	5	After the 5th transaction (ownership ranking- 5) the vehicle will obtain an invalid status and the commodity cannot be further traded.
3.	Hatch Back	3	3	After the 3rd transaction (ownership ranking- 3) the vehicle will obtain an invalid status and the commodity cannot be further traded.
4.	Trucks	3	3	Vehicle will obtain invalid status after 3rd transaction

Depreciation % Table:

SL NO	Vehicle Type	Maximum Number of Transactions allowed	Ownership Ranking that will allot invalid status	Comments
1.	SUV	10	10	After the 10th transaction (ownership ranking- 10) the vehicle will obtain an invalid status and the commodity cannot be further traded.
2.	Sedan	5	5	After the 5th transaction (ownership ranking- 5) the vehicle will obtain an invalid status and the commodity cannot be further traded.
3.	Hatch Back	3	3	After the 3rd transaction (ownership ranking- 3) the vehicle will obtain an invalid status and the commodity cannot be further traded.
4.	Trucks	3	3	Vehicle will obtain invalid status after 3rd transaction

Depreciation % Table:

SUV:

SL NO	Vehicle Type	Ownership Ranking	In-Valid Status	Depreciation Factor (D.F.)			Comments
1.	SUV	I	False	20% (after rolling out from showroom)			Eligible for Trade
2.	SUV	II	False	Distance Driven <10k km: 2% (D.F.)	Distance Driven >10k to 50k km: 3.0% (D.F.)	Distance Driven >=50k km: 3.5% (D.F.)	Eligible for Trade
3.	SUV	III	False	Distance Driven <10k km: 4% (D.F.)	Distance Driven >10k to 50k km: 5.0% (D.F.)	Distance Driven >=50k km: 5.5% (D.F.)	Eligible for Trade
4.	SUV	IV	False	Distance Driven <10k km: 6% (D.F.)	Distance Driven >10k to 50k km: 7.0% (D.F.)	Distance Driven >=50k km: 7.5% (D.F.)	Eligible for Trade
5.	SUV	V	False	Distance Driven <10k km: 8% (D.F.)	Distance Driven >10k to 50k km: 9.0% (D.F.)	Distance Driven >=50k km: 9.5% (D.F.)	Eligible for Trade
6.	SUV	VI	False	Distance Driven <10k km: 10% (D.F.)	Distance Driven >=10k to 50k km: 11.0% (D.F.)	Distance Driven >=50k km: 11.5% (D.F.)	Eligible for Trade
7.	SUV	VII	False	Distance Driven <10k km: 12% (D.F.)	Distance Driven >=10k to 50k km: 13.0% (D.F.)	Distance Driven >=10k to 50k km: 13.5% (D.F.)	Eligible for Trade
8.	SUV	VIII	False	Distance Driven <10k km: 14% (D.F.)	Distance Driven >=10k to 50k km: 15.0% (D.F.)	Distance Driven >=10k to 50k km: 15.5% (D.F.)	Eligible for Trade
9.	SUV	IX	False	Distance Driven <10k km: 16% (D.F.)	Distance Driven >=10k	Distance Driven >=10k to	Eligible for Trade

					to 50k km: 17.0% (D.F.)	50k km: 17.5% (D.F.)	
10.	SUV	X	True	22%			No Longer Eligible

Note: The 10th owner of an SUV vehicle will no longer be able to re-sell his/her car.

Sedan:

SL NO	Vehicle Type	Ownership Ranking	In-Valid Status	Depreciation Factor			Comments
1.	Sedan	I	False	20% (after rolling out from showroom)			Eligible for Trade
2.	Sedan	II	False	Distance Driven <10k km: 4% (D.F.)	Distance Driven >=10k to 50k km: 6.0% (D.F.)	Distance Driven >=50k km: 7.5% (D.F.)	Eligible for Trade
3.	Sedan	III	False	Distance Driven <10k km: 8% (D.F.)	Distance Driven >=10k to 50k km: 9.0% (D.F.)	Distance Driven >=50k km: 10.5% (D.F.)	Eligible for Trade
4.	Sedan	IV	False	Distance Driven <10k km: 12% (D.F.)	Distance Driven >=10k to 50k km: 13.0% (D.F.)	Distance Driven >=50k km: 15% (D.F.)	Eligible for Trade
5.	Sedan	V	True	22%			No Longer Eligible

Note: The 5th owner of a Sedan vehicle will no longer be able to re-sell his/her car.

Hatchback:

SL NO	Vehicle Type	Ownership Ranking	In-Valid Status	Depreciation Factor			Comments
1.	Hatch Back	I	False	20% (after rolling out from showroom)			Eligible for Trade

2.	Hatch Back	II	False	Distance Driven <10k km: 15% (D.F.)	Distance Driven >=10k to 50k km: 17.0% (D.F.)	Distance Driven >=50k km: 18% (D.F.)	Eligible for Trade
3.	Hatch Back	III	True	22%			No Longer Eligible

Note: The **3rd owner of a Hatch Back** vehicle will no longer be able to re-sell his/her car.

Truck:

SL NO	Vehicle Type	Ownership Ranking	In-Valid Status	Depreciation Factor			Comments
1.	Truck	I	False	20% (after rolling out from showroom)			Eligible for Trade
2.	Truck	II	False	Distance Driven <50k km: 20% (D.F.)	Distance Driven >=50k to 100k km: 30.0% (D.F.)	Distance Driven >=100k km: 40% (D.F.)	Eligible for Trade
3.	Truck	III	True	60%			No Longer Eligible

Note: The **3rd owner of a Truck** will no longer be able to re-sell his/her car.

The above *Smart Contract* will prevent black-marketing and hoarding of car prices, prevention of tampering of previous ownership history, distance driven by respective owners and abiding by the car price regulation as decided by Government.

Gaps in Carfax and Kelly Blue Book in comparison to the demonstrated proposed use case:

Carfax is a commercial web-based service that provides vehicular details prior to any transaction or reselling of a car if requested by a new owner. Carfax stores the historical data of cars and light trucks and mainly focusses on the American and Canadian Market. The data retrieval process works on the Vehicular Identification Number (VIN). In turn, once the report is generated and handed over to the requester, Carfax charges a fee for the service.

On the other hand, Kelly Blue Book is a vehicle evaluation and automotive research company that provides a fair market price to used automobiles and motorbikes by taking standard depreciation into account and various parameters during selling and reselling based on the VIN number.

However, the gap that exists for both Carfax and Kelly Blue Book that the data is stored in their respective database, the algorithm is not very open that is used for monitoring the price of used vehicles and a tamper of historical data can take place without even intimating the automotive

sector. The report generated by these organizations are not full proof and cannot be presented in court (Lehto, 2017)

The prototype and the Hyperledger use case will fill the gap through a complete tamper proof historical vehicular system.

Gaps in the Literature

The literature review of various applications of Blockchain Technology show about the benefits and the possible disruption that it might create due to technological advancements and breakthrough. However, most of the improvements in the technological up-gradations have been made by seeing the end results without considering the transition phase. In most of the sectors, when Blockchain gets incorporated as a backbone of a business process, it would pose significant challenges as there would be a technological gap between the people who were accustomed to use application services in an older technological environment. On reading most of the research articles, this issue has not been covered however it would create a significant impact during the transition period. Also most of the places, do not have a faster access to the internet that is creating a challenge for the Blockchain Technology to be considered as a potential to be a game changer.

Concept Matrix of the reviewed literature

Articles	Concepts															
	Transportation				Business				IOT				Healthcare			
Unit of Analysis	T	P	C	LF	T	P	C	LF	T	P	C	LF	T	P	C	LF
Nayak, A., & Dutta, K. (2017)					1		1					1				
Zyskind, G., Nathan, O., & Pentland, A. S. (2015).						1		1	1		1					
Mettler, M. (2016).								1					1			1
Zhang, J., Xue, N., Huang, X. (2017).											1	1	1		1	1
Yuan, Y., & Wang, F. Y. (2016).	1		1	1			1				1	1				

Dorri, A., Steger, M., Kanhere, S. S., & Jurdak, R. (2017).	1	1	1	1	1		1			1						
Biswas, K., & Muthukkumarasamy, V. (2016).					1		1		1		1					
Dorri, A., Kanhere, S. S., Jurdak, R., & Gauravaram, P. (2017).					1		1		1	1	1					
Christidis, K., & Devetsikiotis, M. (2016).					1	1			1	1	1	1				
Deloitte. (2018).					1	1	1	1								
Petek, I., & Zajec, N. (2018).													1		1	
Sousa, J., Bessani, A., & Vukolic, M. (2018).					1		1									
Cha, S. C., Peng, W.C., Hsu, T. Y., Chang, Chu. Lin., & Li, S. W. (2018).						1		1								
Johng, H., Kim, D., Hill, T., & Chung, L. (2018).					1		1	1								
Kaid, D., & Eljazzar, M, M. (2018).					1	1	1									
Yang, J., Lu, Z., & We, J. (2018)					1	1		1	1	1		1				
Ali, J., Ali, T., Musa, S., Zahrani, S. (2018).									1		1					
Biswas, K., & Muthukkumarasamy, V. (2016)				1				1				1				

Linux Foundations. (2018)					1	1		1	1		1	1				
Total	2	1	2	3	11	7	9	8	8	4	8	7	3	0	2	2

Legend: T (Trust), P (Payments), C (Cryptography), LF (Low Friction)

Key Takeaways

Financial services would benefit the key advantages of Blockchain with the acceptance and popularity of crypto-currency (Kaid & Eljazzar, 2018).

Healthcare- Blockchain has the capability to transform the treatment of patients and increase the health expectancy of the people specially in developing and under-developed nation (Mettler, 2016).

Complete tamper proof, irrevocable and high safety of data management would draw significant research to take place as the model of Blockchain technology can fit in almost every key application areas (Yang et al ., 2018).

Conclusion

At present, Blockchain Technology is still more focused in the research phase with very little up and running applications in today's world. The disruptive character that underlies this new kind of technology would strongly affect the balance of power between existing market players. It has also the potential to promote new kind of digital business models and industrial initiatives and be self-rewarding in nature with days to come. Looking at the possible functionalities of Blockchain technology, it would open the doors to different kinds of new practice and spread its application in a wide variety of sectors and bring along significant disruption with it.

The gaps that have been identified in the literature review would be the stepping stone to focus and form the key areas of research. Future research can also be carried out to study the key technical requirements so as to fit in Blockchain seamlessly rather than creating a complete overhaul in the current business practices. The study that has been done showcases, "Hyperledger" as a top-enterprise of Blockchain technology and has the potential of high scalability of different kinds of businesses at present thus creating a research opportunity in terms of key applications of blockchain.

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